



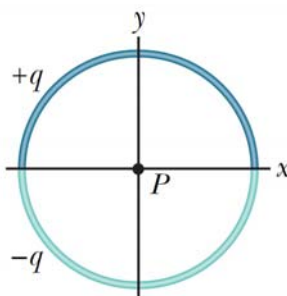
南方科技大学  
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: Principles of physics IIDept.: PhysicsExam Duration: 2 hoursExam Paper Setter: Physics teaching team

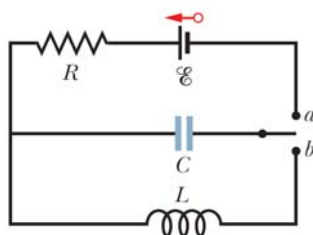
| Question No. | 1  | 2  | 3  | 4  | 5  |  |  |  |  |  |
|--------------|----|----|----|----|----|--|--|--|--|--|
| Score        | 20 | 20 | 20 | 20 | 20 |  |  |  |  |  |

### Comprehensive questions

**Q2.** Two curved plastic rods, one of charge  $+q$  and the other of charge  $-q$ , form a circle of radius  $R = 8.50$  cm in an  $x$ - $y$  plane. The  $x$  axis passes through both of the connecting points, and the charge is distributed uniformly on both rods. If  $q = 15.0$  pC, what are the (a) magnitude and (b) direction of the electric field at  $P$ , the center of the circle?

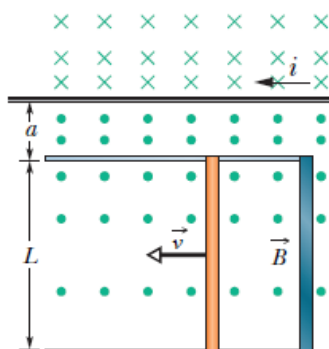


**Q3.** In the figure,  $R = 14.0 \Omega$ ,  $C = 31.2 \mu\text{F}$ , and  $L = 54.0$  mH, and the ideal battery has emf  $\mathcal{E} = 34.0$  V. The switch is kept at  $a$  for a long time and then thrown to position  $b$ . What are the (a) frequency and (b) current amplitude of the resulting oscillations?



**Q4.** The figure shows a rod of length  $L=10.0$  cm that is forced to move at constant speed  $v=5.00$  m/s along horizontal rails. The rod, rails, and connecting strip at the right form a conducting loop. The rod has resistance  $0.40$ ; the rest of the loop has negligible resistance. A current  $i=100$  A through the long straight wire at distance  $a=10.0$  mm from the loop sets up a (nonuniform) magnetic field through the loop. Find:

- The emf and current induced in the loop.
- At what rate is thermal energy generated in the rod?
- What is the magnitude of the force that must be applied to the rod to make it move at constant speed?



**Q5.** Light of wavelength  $550$  nm is incident normally on a double slits. Two adjacent maxima occur at angles given by  $\sin \theta = 0.2$  and  $\sin \theta = 0.3$ . The fourth-order maxima are missing.

- What is the separation between adjacent slits?
- What is the smallest slit width this slit can have?
- For that slit width, what are the largest values of the order number  $m$  of the maxima produced by the slits?

**Q6.** A space traveler takes off from Earth and moves at speed  $0.9950c$  toward the star Vega, which is  $26.00$  ly distant. How much time will have elapsed by Earth clocks (a) when the traveler reaches Vega and (b) when Earth observers receive word from the traveler that she has arrived? (c) How much older will Earth observers calculate the traveler to be (measured from her frame) when she reaches Vega than she was when she started the trip?